Smart Image Processing & AR Filter System using Computer Vision

ABSTRACT

This project titled **"Smart Image Processing & AR Filter System using Computer Vision"** focuses on applying various computer vision techniques to images using a web-based interface. The system allows users to input an image URL and apply different filters such as **blur, edge detection, contour detection, and grayscale processing**. Additionally, the system supports **Augmented Reality (AR) filters**, allowing fun visual overlays like sunglasses, hats, or other facial accessories.

The project is implemented using **Python, OpenCV, and Streamlit**, providing a simple and interactive web application. Users can upload or provide an image URL, select a filter operation, and view the processed output instantly.

This project demonstrates practical applications of **computer vision techniques in real-time image processing**, making it useful for learning, experimentation, and multimedia applications.

INTRODUCTION

Computer Vision is a field of Artificial Intelligence that enables computers to interpret and process visual information from images and videos. Image processing techniques are widely used in applications such as facial recognition, surveillance systems, augmented reality, and image editing tools.

This project integrates **image filtering and augmented reality features** into a single web application. Using **OpenCV**, the system processes images with various filters, while **Streamlit** provides an interactive interface where users can select operations and visualize results.

The system demonstrates how computer vision techniques can be used to manipulate images efficiently and interactively through a simple web-based platform.

OBJECTIVES

The main objectives of this project are:

- To develop a web-based image processing system
- To apply different computer vision filters to images
- To allow users to input image URLs for processing
- To implement image transformations like blur, edge detection, and contour detection
- To apply AR filters such as sunglasses or hats on images
- To display processed images instantly on the interface
- To create an easy-to-use interface using Streamlit

SYSTEM REQUIREMENTS

Hardware Requirements

• Computer or Laptop
• Minimum 4 GB RAM
• Internet connection (for image URL input)

Software Requirements

• Operating System: Windows / Linux / macOS
• Python 3.8 or above
• Streamlit
• OpenCV
• NumPy
• Pillow
• Requests

TECHNOLOGIES USED

Python

Python is used as the main programming language due to its simplicity and extensive support for machine learning and computer vision libraries.

OpenCV

OpenCV is used for performing image processing tasks such as:

• Reading and displaying images
• Applying filters like blur and edge detection
• Detecting contours in images
• Implementing augmented reality overlays

Streamlit

Streamlit is used to create the web interface and handle user interaction.

It allows:
• Image display
• Filter selection
• Interactive UI components

NumPy

NumPy is used for efficient array operations and image data manipulation.

MODULE DESCRIPTION

Image Input Module

• Accepts image URL from the user
• Downloads and converts the image into a format suitable for processing

Image Processing Module

• Applies selected filters such as:

- Blur filter
- Edge detection
- Contour detection
- Grayscale conversion

AR Filter Module

• Applies overlay images such as:

- Sunglasses
- Hat
- Crown
  • Demonstrates basic augmented reality effects

User Interface Module

• Built using Streamlit
• Provides filter selection options
• Displays original and processed images

WORKING PRINCIPLE

1. The user enters an image URL into the system
2. The image is downloaded and converted into a NumPy array
3. The user selects an image filter or AR effect
4. OpenCV processes the image using the selected technique
5. The processed image is displayed on the Streamlit interface

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ALGORITHM

1. Start the application
2. Accept image URL from the user
3. Load the image using requests and OpenCV
4. Convert image into array format
5. Apply selected image filter
6. Display processed image on screen
7. Allow user to select different filters

ADVANTAGES

- Simple and user-friendly interface
- Supports multiple image processing techniques
- Demonstrates real-time computer vision operations
- Useful for learning OpenCV concepts
- Can be extended with more filters

APPLICATIONS

- Image editing tools
- Augmented reality applications
- Educational demonstrations
- Social media filter systems
- Multimedia processing systems

LIMITATIONS

- Works only with valid image URLs
- AR filters are basic overlays
- Performance depends on internet speed and system hardware

FUTURE ENHANCEMENTS

- Real-time webcam filters
- Face detection-based AR filters
- Upload image feature instead of URL only
- AI-based object detection
- Mobile application integration

CONCLUSION

The **Smart Image Processing & AR Filter System** successfully demonstrates how computer vision techniques can be applied to images using Python and OpenCV. By integrating Streamlit, the system provides an interactive and easy-to-use platform where users can apply filters and view processed results instantly.

This project serves as a practical introduction to **image processing and augmented reality concepts**, and it can be extended further with advanced AI and deep learning features.